

Team Profiles

Archit Naraniwal

CO-FOUNDER, FACLON LABS

10+ years of experience working with industrial companies on data, technology and AI.

- ▮ Built Faclon Labs from the ground up, focused on solving real operational problems for manufacturing and industrial companies — now 50+ enterprise customers and 200+ deployments.
- ▮ Leads the Cement vertical at Faclon Labs, one of the company's key focus sectors alongside steel, metals and power — owning the cement offering across pyro, grinding, energy and reliability.
- ▮ Leads the cement business end-to-end — from identifying customer problems and high-value opportunities to solution design, customer engagement, deployment and value realisation.
- ▮ Works closely with leading industrial groups including JSW Group, Vedanta Group, ITC, Adani Group, Nuvoco and Dalmia Bharat — at plant and corporate level, from first pilot to multi-site rollout.
- ▮ Brings technology and business together, with an IIT Bombay background, prior experience at PwC and years of building an industrial technology company alongside plant teams.

Vivek Singh

CHIEF AI OFFICER, FACLON LABS

Heads Faclon's AI and data-science team — the models behind its cement deployments in preheater, kiln and grinding, and its metals and power-plant work. IIT Kanpur, BS-MS Mathematics & Scientific Computing; ex-Blue Yonder, Delhivery.

- ▮ Built Faclon's cyclone build-up severity model — differential pressure, temperatures, draft, O₂, raw-mix and fuel chemistry and kiln load combined into a live score that flags build-up 30-60 minutes ahead, with likely cause.
- ▮ Built the kiln models at JSW Cement Nandyal — kiln and coal mill fuel optimisation with operator-approved recommendations, alternative-fuel ratio optimisation — and the kiln heat-balance and SHC set-point application.
- ▮ Built the grinding analytics — root-cause analysis of high specific power in JSW Nandyal's raw and cement mills, kiln and mill operator scoring, and the grinding circuit optimiser (Blaine and residue via separator advice).
- ▮ Also built clinker C3S and strength prediction from kiln and raw-mix data, the per-kiln decision engine for MSP Steel's sponge-iron (DRI) kilns, and net-heat-rate models for captive power plants at Hindustan Zinc and Grasim.
- ▮ Leads development of Faclon's AI stack — DeepSense (model library, MLOps) and Bruce AI (agents with human-in-the-loop checks and guardrails) — used across 30+ productised industrial applications.

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Vijay Kumar Komaragiri

CEMENT PROCESS SME - ADVISOR, FACLON LABS

30+ years in cement with ACC, Blue Circle, Lafarge and LafargeHolcim in India and Yanbu Cement in Saudi Arabia — from process engineering, projects and technical support to business-unit and country leadership. M.Tech, IIT Kharagpur.

- ▮ Plant process experience across pyro (preheater, kiln, cooler) and grinding (raw and cement mills) — the areas behind the cyclone jamming, kiln heat balance and ball mill use cases — with a focus on process optimisation.
- ▮ On Faclon's cement AI programmes, defines and validates the process mechanisms and signatures the models must capture — cyclone build-up, kiln heat balance, mill stability — and flags what cannot be predicted.
- ▮ Leads root-cause analysis, operating recommendations and SME workshops with plant operations teams, so that model outputs are checked against plant practice.
- ▮ Led the implementation of the global LafargeHolcim Cement Industrial Operational Framework across 15 ACC plants and its integration with talent, performance and career-path management.
- ▮ Responsible for identifying and implementing digitisation initiatives with LafargeHolcim's regional technical and IT centres — plant-side experience of what digital tools must do to be adopted by operations teams.

Select credentials demonstrating experience in cyclone jamming prediction, kiln heat balance / SHC optimisation and mill grinding optimisation



JSW Cement - Preheater & kiln

- ▶ Cyclone build-up severity index – 30–60 min early warning with corrective actions
- ▶ Upsets down from 6 to 2 a year; availability +1.5%; ₹1.86 Cr/yr saved
- ▶ Kiln & coal mill fuel optimisation (operator-approved set-points); kiln SPC down ~10%
- ▶ C3S prediction – variability down ~16%; kiln MTBF up ~9.5x



JSW Cement - Mills & energy

- ▶ Specific-power root-cause analysis for raw and cement mills – RCA in hours, not days
- ▶ Raw mill: SPC down 9–10%, maintenance cost down 52%, MTBF up 5–7x
- ▶ Plant energy platform on DCS, meter, lab and SAP data – ₹1.2–2.7 Cr/yr energy saved
- ▶ AI demand forecasting for power booking; ₹4–5 Cr/yr plant savings



UltraTech Cement

- ▶ Plant energy & PowerMix management across 13 plants on one template – 800+ meters
- ▶ Busbar thermal monitoring on 47 critical HT joints – overheat caught before failure
- ▶ Digital Logbook & Safety at Awarpur – 300 users, 15 departments, 1,000+ incidents
- ▶ Solar generation monitoring across the cluster



Grasim (Aditya Birla Group)

- ▶ Energy management across 4 plants on the group's cloud – 2,500+ meters, 200+ users
- ▶ AI benchmarking and SEC prediction – ₹10 Cr savings identified, payback under 1 year
- ▶ Steam-trap, vibration (100+), VFD (200+) and busbar (100+) health monitoring
- ▶ AI net-heat-rate reduction for the 175 MW captive power plant (in progress)



Dalmia Bharat Cement

- ▶ Multi-plant energy management system – 300 energy meters on one platform
- ▶ Energy accountability down to PCC and equipment level



MSP Steel - DRI kilns

- ▶ Kiln AI model and digital twin on DCS, lab and asset-health data
- ▶ Live stability, efficiency and quality indices for every kiln
- ▶ Per-kiln decision engine – Increase / Maintain / Reduce guidance for operators
- ▶ Early alerts for accretion risk – longer campaign life, fewer unplanned stoppages



Hindustan Zinc

- ▶ Zawar 91 MW captive power plant – AI for auxiliary power and net heat rate (target 1 pp cut)
- ▶ Cell-house monitoring across 66 cells; scaling to 330 cells across 3 plants
- ▶ Vision AI on three conveyor systems – fault, misalignment and spillage detection



JSW Energy

- ▶ Predictive maintenance across five power plants – 1,000+ vibration sensors
- ▶ 15+ critical failures detected early; less unplanned downtime and maintenance cost
- ▶ ₹1.3 Cr savings over two years, validated by the client's finance team